

SATHISH KANTHI

AI Engineer · Generative AI, RAG & Agentic AI Applications

+91 7659876245 | Kanthi.sathish777@gmail.com | [LinkedIn](#) | [GitHub](#) | Hyderabad, India

PROFESSIONAL SUMMARY

AI Engineer specializing in Generative AI, Retrieval-Augmented Generation (RAG), and Agentic AI applications. Experienced in building LLM-powered software using Python, Lang-Chain, vector databases, and prompt engineering to solve real-world business problems. Hands-on experience developing multi-step AI agents, retrieval pipelines, semantic search systems, and production-oriented AI applications. B.Tech in Artificial Intelligence (CGPA 8.19, First Class Honours) with strong software engineering fundamentals with a focus on rapidly building, testing, deploying, and improving AI products.

WORK EXPERIENCE

Artificial Intelligence Engineer Intern — iPlairani.AI

June 2026 – Present

- Built an AI Knowledge Application using Retrieval-Augmented Generation (RAG) and Agentic AI workflows for intelligent enterprise document search.
- Developed multi-step AI agents capable of reasoning, tool calling, and context-aware retrieval using Lang-Chain.
- Designed document chunking, embedding generation, and vector retrieval pipelines to improve retrieval accuracy and reduce hallucinations.
- Engineered prompts and retrieval strategies to improve answer quality across different enterprise knowledge sources.
- Built semantic search pipelines integrating SQL data and vector databases for efficient document retrieval.
- Optimized inference performance using INT8/INT4 quantization techniques while maintaining response quality.
- Explored robotics AI integration concepts, including sensor fusion and AI-driven decision pipelines.
- Worked collaboratively using Git for version control and iterative feature development.

Technologies: Python, LangChain, RAG, LLMs, Prompt Engineering, Vector Databases, SQL, Quantization, Git

Machine Learning Intern — Fly-Rank.AI

July 2026 – Present

- Building NLP pipelines for semantic search, keyword extraction, and content relevance scoring as part of the Applied Search Intelligence initiative.
- Developing machine learning solutions for search ranking optimization using large-scale search datasets.
- Researching SERP feature optimization, including featured snippets, knowledge panels, and structured data markup strategies.
- Processing and analyzing structured and unstructured data to improve content discoverability.
- Building Python data pipelines for search intelligence workflows and feature engineering.
- Collaborating with engineering teams using Git and Python to develop production-ready ML solutions.

Technologies: Python, NLP, Scikit-learn, Search APIs, Data Pipelines, Git

SELECTED AI PROJECTS

Real-Time Emotion Detection System — Dual Modal Input

Nov 2024 – Feb 2025

- Developed a dual-modal deep learning system combining facial expression recognition and speech emotion detection.
- Built a CNN-based facial emotion recognition model achieving ~85% accuracy, with confidence-weighted late fusion across modalities.
- Designed an MFCC-based speech feature extraction pipeline using Librosa for live audio streams.
- Integrated OpenCV for real-time video capture and inference, with a Tkinter GUI for live visualization.

Technologies: Python, TensorFlow, OpenCV, Librosa, CNN, RNN, Tkinter

Intelligent Fraud Detection System Using Machine Learning

Mar 2023 – May 2023

- Built an end-to-end ML pipeline for detecting fraudulent transactions on highly imbalanced financial datasets.
- Performed EDA, feature engineering, and class imbalance handling (SMOTE, class-weight tuning).
- Compared Logistic Regression, Random Forest, and XGBoost, evaluated on precision, recall, and F1 given the business cost of false negatives.
- Designed visual analytics dashboards showing fraud patterns by transaction type, amount, and time.

Technologies: Python, Pandas, scikit-learn, NumPy, Matplotlib, MySQL

EDUCATION

Vidya Jyothi Institute of Technology, Hyderabad

2021 – 2025

B.Tech in Artificial Intelligence · CGPA: 8.19 / 10.00 · First Class with Honours

- Final-year project: Real-Time Emotional Analysis with Dual Modal Input (CNN + RNN, ~85% accuracy).
- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, DBMS, Probability & Statistics.

Nano Junior College, Hyderabad

2018 – 2020

Intermediate — MPC · 86.3%

TECHNICAL SKILLS

Programming: Python, SQL, Java

AI Engineering: LLMs, Agentic AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, LangChain, Semantic Search, Embeddings, Vector Databases, Tool Calling

Machine Learning: TensorFlow, Keras, scikit-learn, OpenCV, Librosa

Databases: SQL, MySQL, Vector Databases (ChromaDB, Pinecone)

Software Engineering: Git, GitHub, VS Code, PyCharm, Jupyter Notebook

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn, Power BI

Concepts: Deep Learning, NLP, Computer Vision, Quantization, Big Data

CERTIFICATIONS

- Building LLM Applications Using Prompt Engineering — NVIDIA
- Machine Learning — AWS Academy
- Data Engineering — AWS Academy
- Neural Network and Deep Learning — GreatLearning
- AI Foundation & Advanced — Hexart.ai
- Introduction to C Programming — IIT Bombay
- Python Basics — HackerRank

LANGUAGES

English · Telugu · Hindi